

Understanding ELF, the Executable and Linkable Format

Whenever we compile any code, the output that we get is an executable file, which we generally don't bother about. We execute it on our desired target. If it is an embedded system, we flash it to the target device and run it. Usually, this executable file is the ELF (Executable and Linkable Format) file. In this article we will analyse the compilation process as well as ELF.



To begin with, there are many compilation stages so let us look at them more closely.

Compilation stages

To understand the stages of compilation, let us write a simple program and compile it. For compilation purposes, we will use the GCC open source compiler. So, let us consider a file *study.c* that contains the following program:

```
#include <stdio.h>
#define NUMBER_1 1
#define NUMBER_2 2
```

```
int main(void)
{
    int a,b,c;
    a = NUMBER_1;
    b = NUMBER_2;
    c = a + b;

    printf("Output: %d\r\n", c);

    return 0;
}
```

We use the following command to compile this *study.c* programming file:

```
gcc study.c -o study
```

Note: GCC or other tools mentioned in this article need to be installed before they are used.

The above command compiles 'study.c' and produces an executable file, 'study'. Type the following command to see the file type of this executable:

```
file study
Output:
```

```
study: ELF 64-bit LSB shared object,
x86-64
```